



The Mismatched Message

Quick facilitation notes & answer key

Goal

Students debug a broken EVENT link in concurrent code (C3.2 read/alter + C3.1 write). Pixel never starts because its receive hat ('when I receive stop') does not match Bit's broadcast ('go'). Students compare the two message words, see they differ, and fix one so they match. Debugging message mismatches is a key reading skill for event-driven concurrent code.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two message cards reading 'go' and 'stop'; hold them side by side to show they do not match, then change one.

Run it (about 20 min)

1. I DO: run it — 'Bit goes 0 forward 1 → 1 and broadcasts go. But Pixel is waiting for stop, so it never hears its message.'
2. WE DO: compare the words — 'go' vs 'stop'. Different, so Pixel's hat never fires; it stays at 5.
3. YOU DO: students fix one message so both say the same word, then run Pixel: 5 back 2 → 3.
4. Connect: 'A receive hat needs the EXACT message it waits for — check the words match.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 1 → 1. Pixel does NOT move (circle NO); it stays on its home number 5.

Ex 2 — Bit broadcasts 'go'; Pixel waits for 'stop'. They are NOT the same word, so Pixel's receive event never fires.

Ex 3 — fix: change Pixel's hat to 'when I receive go' (or Bit's broadcast to 'stop'). With matching messages, Pixel runs: 5 → back 2 → 3.

Fix note: either edit works as long as BOTH messages end up the same word; then Pixel reaches its goal 3.

Watch for & differentiate

Common error: changing a move block — the bug is the EVENT message, not Pixel's motion.

Simplify: have students circle the two message words and check letter by letter if they match.

Extend: ask what happens if Bit's 'broadcast go' block is deleted (Pixel never starts no matter what its hat says).

Success indicator: the child reports Pixel stuck at 5, names the message mismatch, and fixes a message so Pixel reaches 3.